# Static Analysis

## Bandit:

- **Before:**

```
┌──(kali㉿kali)-[/media/sf_Shared]
└─$ bandit -r ./ShushChat_v1/
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.9
Run started:2026-05-02 14:39:34.663965

Test results:
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.7.10/plugins/b110_try_except_pass.html
   Location: ./ShushChat_v1/gui.py:678:16
677                     self._net.send_to(pid, update_msg)
678                 except Exception:
679                     pass
680

--------------------------------------------------
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.7.10/plugins/b110_try_except_pass.html
   Location: ./ShushChat_v1/gui.py:803:12
802                             return
803                 except Exception:
804                     pass
805                 self._handle_encrypted(peer_id, raw)

--------------------------------------------------
>> Issue: [B112:try_except_continue] Try, Except, Continue detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.7.10/plugins/b112_try_except_continue.html
   Location: ./ShushChat_v1/gui.py:928:12
927                     port = entry.get("port")
928                 except Exception:
929                     continue
930
```

```
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.7.10/plugins/b110_try_except_pass.html
   Location: ./ShushChat_v1/gui.py:1008:12
1007                    self._net.send_to(peer_id, enc)
1008                except Exception:
1009                    pass
1010
--------------------------------------------------
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.7.10/plugins/b110_try_except_pass.html
   Location: ./ShushChat_v1/gui.py:1141:12
1140                    self._net.send_to(pid, enc)
1141                except Exception:
1142                    pass
1143
--------------------------------------------------
>> Issue: [B104:hardcoded_bind_all_interfaces] Possible binding to all interfaces.
   Severity: Medium   Confidence: Medium
   CWE: CWE-605 (https://cwe.mitre.org/data/definitions/605.html)
   More Info: https://bandit.readthedocs.io/en/1.7.10/plugins/b104_hardcoded_bind_all_interfaces.html
   Location: ./ShushChat_v1/gui.py:1237:30
1236                    bind_ip = self._ip_entry.text().strip()
1237                    if bind_ip == "0.0.0.0" or not bind_ip:
1238                        my_ips = self._get_local_ips()
--------------------------------------------------
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.7.10/plugins/b110_try_except_pass.html
   Location: ./ShushChat_v1/gui.py:1335:8
1334                self._net.disconnect_all()
1335            except Exception:
1336                pass
1337            if self._group:
```

```
--------------------------------------------------
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.7.10/plugins/b110_try_except_pass.html
   Location: ./ShushChat_v1/main.py:52:4
51          windll.shcore.SetProcessDpiAwareness(1)
52      except Exception:
53          pass
54  window = ShushChatApp()
--------------------------------------------------
>> Issue: [B104:hardcoded_bind_all_interfaces] Possible binding to all interfaces.
   Severity: Medium   Confidence: Medium
   CWE: CWE-605 (https://cwe.mitre.org/data/definitions/605.html)
   More Info: https://bandit.readthedocs.io/en/1.7.10/plugins/b104_hardcoded_bind_all_interfaces.html
   Location: ./ShushChat_v1/networking.py:337:22
336                 srv.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
337                 srv.bind(("0.0.0.0", port))
338                 srv.listen(16)
--------------------------------------------------

Code scanned:
        Total lines of code: 1911
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 7
                Medium: 2
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 2
                High: 7
Files skipped (0):
```

- **After:**

```
┌──(kali㉿kali)-[/media/sf_Shared/ShushChat_OBF/ShushChat_v1_Final/ShushChat_v1]
└─$ bandit -r .
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.9
Run started:2026-05-04 19:06:30.043484

Test results:
>> Issue: [B104:hardcoded_bind_all_interfaces] Possible binding to all interfaces.
   Severity: Medium   Confidence: Medium
   CWE: CWE-605 (https://cwe.mitre.org/data/definitions/605.html)
   More Info: https://bandit.readthedocs.io/en/1.7.10/plugins/b104_hardcoded_bind_all_interfaces.html
   Location: ./gui.py:1257:30
1256                    bind_ip = self._ip_entry.text().strip()
1257                    if bind_ip == "0.0.0.0" or not bind_ip:
1258                        my_ips = self._get_local_ips()

--------------------------------------------------


Code scanned:
        Total lines of code: 1984
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 1
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 1
                High: 0
Files skipped (0):
```

## Pylint:

- **Before:**

```
networking.py:140: R0801: Similar lines in 2 files
==gui:[101:108]
==networking:[120:129]
    try:
        ipaddress.ip_address(ip)
        return True
    except ValueError:
        return False


def _is_valid_port(port: int) -> bool:
    """Return True if *port* is in the valid TCP port range."""  (duplicate-code)

------------------------------------------------------------------
Your code has been rated at 7.65/10 (previous run: 7.65/10, +0.00)
```

- **After:**

```
************* Module networking
networking.py:28:12: C3001: Lambda expression assigned to a variable. Define a function using the "def" keyword
networking.py:28:20: R1728: Consider using a generator instead 'sum(1 for _ in range(4))' (consider-using-genera
networking.py:138:0: R0903: Too few public methods (1/2) (too-few-public-methods)
networking.py:183:0: R0902: Too many instance attributes (9/7) (too-many-instance-attributes)
networking.py:186:4: R0913: Too many arguments (6/5) (too-many-arguments)
networking.py:186:4: R0917: Too many positional arguments (6/5) (too-many-positional-arguments)
networking.py:206:31: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
networking.py:225:4: C0116: Missing function or method docstring (missing-function-docstring)
networking.py:266:32: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
networking.py:268:4: C0116: Missing function or method docstring (missing-function-docstring)
networking.py:288:4: C0116: Missing function or method docstring (missing-function-docstring)
networking.py:292:0: R0902: Too many instance attributes (11/7) (too-many-instance-attributes)
networking.py:326:4: C0116: Missing function or method docstring (missing-function-docstring)
networking.py:329:4: C0116: Missing function or method docstring (missing-function-docstring)
networking.py:337:17: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
networking.py:350:16: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
networking.py:393:4: C0116: Missing function or method docstring (missing-function-docstring)
networking.py:414:17: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
networking.py:434:16: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
networking.py:456:4: C0116: Missing function or method docstring (missing-function-docstring)
networking.py:468:4: C0116: Missing function or method docstring (missing-function-docstring)
networking.py:479:4: C0116: Missing function or method docstring (missing-function-docstring)
networking.py:483:4: C0116: Missing function or method docstring (missing-function-docstring)
networking.py:487:4: C0116: Missing function or method docstring (missing-function-docstring)
networking.py:496:4: C0116: Missing function or method docstring (missing-function-docstring)
networking.py:506:4: C0116: Missing function or method docstring (missing-function-docstring)
networking.py:512:4: C0116: Missing function or method docstring (missing-function-docstring)
networking.py:524:7: W0125: Using a conditional statement with a constant value (using-constant-test)

------------------------------------------------------------------
Your code has been rated at 8.08/10 (previous run: 8.05/10, +0.03)
```

## Radon:

- **Before:**

```
┌──(kali㉿kali)-[/media/sf_Shared/ShushChat_v1]
└─$ radon cc -s .
crypto.py
    M 193:4 Session.decrypt - C (14)
    M 86:4 KeyPair.derive_shared_key - A (5)
    C 131:0 Session - A (5)
    M 389:4 GroupSession.mark_seen - A (5)
    F 286:0 peer_fingerprint_from_bytes - A (3)
    C 67:0 KeyPair - A (3)
    M 144:4 Session.__init__ - A (3)
    M 161:4 Session.encrypt - A (3)
    C 299:0 GroupSession - A (3)
    M 324:4 GroupSession.add_peer - A (3)
    M 384:4 GroupSession.is_duplicate - A (3)
    F 52:0 _zero_bytes - A (2)
    M 81:4 KeyPair.fingerprint - A (2)
    M 347:4 GroupSession.remove_peer - A (2)
    M 357:4 GroupSession.peer_fingerprint - A (2)
    M 368:4 GroupSession.encrypt - A (2)
    M 375:4 GroupSession.decrypt - A (2)
    M 402:4 GroupSession.clear_all - A (2)
    M 70:4 KeyPair.__init__ - A (1)
    M 74:4 KeyPair.public_key_bytes - A (1)
    M 121:4 KeyPair.clear - A (1)
    M 268:4 Session._hmac - A (1)
    M 273:4 Session.clear - A (1)
    M 315:4 GroupSession.__init__ - A (1)
    M 353:4 GroupSession.has_peer - A (1)
    M 362:4 GroupSession.active_peer_ids - A (1)
gui.py
    M 833:4 ShushChatApp._handle_handshake - C (20)
    M 952:4 ShushChatApp._handle_encrypted - C (15)
    M 908:4 ShushChatApp._rx_peer_list - C (14)
    M 1027:4 ShushChatApp._rx_chat - C (14)
    M 515:4 ShushChatApp._refresh_room_list - B (10)
    M 1074:4 ShushChatApp._rx_room_hello - B (10)
    M 551:4 ShushChatApp._switch_room - B (9)
    M 1254:4 ShushChatApp._rx_conn_req - B (8)
    M 1344:4 _RoomDialog.__init__ - B (8)
    M 755:4 ShushChatApp._broadcast_encrypted - B (7)
    M 1227:4 ShushChatApp._handle_chat_link - B (7)
    M 1275:4 ShushChatApp._show_invite_dialog - B (7)
    M 132:4 RoomWidget.__init__ - B (6)
    M 659:4 ShushChatApp._do_set_username - B (6)
    M 681:4 ShushChatApp._do_connect - B (6)
    M 727:4 ShushChatApp._do_send - B (6)
    M 789:4 ShushChatApp._ui_dispatch - B (6)
    C 194:0 ShushChatApp - A (5)
    M 608:4 ShushChatApp._do_join_room - A (5)
    M 989:4 ShushChatApp._rx_room_join - A (5)
    C 1343:0 _RoomDialog - A (5)
```

```
┌──(kali㉿kali)-[/media/sf_Shared/ShushChat_v1]
└─$ radon cc -s .
crypto.py
    M 193:4 Session.decrypt - C (14)
    M 86:4 KeyPair.derive_shared_key - A (5)
    C 131:0 Session - A (5)
    M 389:4 GroupSession.mark_seen - A (5)
    F 286:0 peer_fingerprint_from_bytes - A (3)
    C 67:0 KeyPair - A (3)
    M 144:4 Session.__init__ - A (3)
    M 161:4 Session.encrypt - A (3)
    C 299:0 GroupSession - A (3)
    M 324:4 GroupSession.add_peer - A (3)
    M 384:4 GroupSession.is_duplicate - A (3)
    F 52:0 _zero_bytes - A (2)
    M 81:4 KeyPair.fingerprint - A (2)
    M 347:4 GroupSession.remove_peer - A (2)
    M 357:4 GroupSession.peer_fingerprint - A (2)
    M 368:4 GroupSession.encrypt - A (2)
    M 375:4 GroupSession.decrypt - A (2)
    M 402:4 GroupSession.clear_all - A (2)
    M 70:4 KeyPair.__init__ - A (1)
    M 74:4 KeyPair.public_key_bytes - A (1)
    M 121:4 KeyPair.clear - A (1)
    M 268:4 Session._hmac - A (1)
    M 273:4 Session.clear - A (1)
    M 315:4 GroupSession.__init__ - A (1)
    M 353:4 GroupSession.has_peer - A (1)
    M 362:4 GroupSession.active_peer_ids - A (1)
gui.py
    M 833:4 ShushChatApp._handle_handshake - C (20)
    M 952:4 ShushChatApp._handle_encrypted - C (15)
    M 908:4 ShushChatApp._rx_peer_list - C (14)
    M 1027:4 ShushChatApp._rx_chat - C (14)
    M 515:4 ShushChatApp._refresh_room_list - B (10)
    M 1074:4 ShushChatApp._rx_room_hello - B (10)
    M 551:4 ShushChatApp._switch_room - B (9)
    M 1254:4 ShushChatApp._rx_conn_req - B (8)
    M 1344:4 _RoomDialog.__init__ - B (8)
    M 755:4 ShushChatApp._broadcast_encrypted - B (7)
    M 1227:4 ShushChatApp._handle_chat_link - B (7)
    M 1275:4 ShushChatApp._show_invite_dialog - B (7)
    M 132:4 RoomWidget.__init__ - B (6)
    M 659:4 ShushChatApp._do_set_username - B (6)
    M 681:4 ShushChatApp._do_connect - B (6)
    M 727:4 ShushChatApp._do_send - B (6)
    M 789:4 ShushChatApp._ui_dispatch - B (6)
    C 194:0 ShushChatApp - A (5)
    M 608:4 ShushChatApp._do_join_room - A (5)
    M 989:4 ShushChatApp._rx_room_join - A (5)
    C 1343:0 _RoomDialog - A (5)
```

```
    M 775:4 ShushChatApp._ui_peer_connected - A (1)
    M 786:4 ShushChatApp._on_raw_message - A (1)
    M 807:4 ShushChatApp._on_peer_disconnected - A (1)
    M 825:4 ShushChatApp._on_network_error - A (1)
    M 1144:4 ShushChatApp._set_conn_badge - A (1)
    M 1154:4 ShushChatApp._set_status - A (1)
    M 1157:4 ShushChatApp._unlock_input - A (1)
    M 1163:4 ShushChatApp._lock_input - A (1)
    M 1186:4 ShushChatApp._now - A (1)
    M 1221:4 ShushChatApp._add_err_to_active - A (1)
main.py
    F 47:0 main - A (2)
networking.py
    M 333:4 MultiPeerManager._listen_loop - B (9)
    M 225:4 PeerConnection._recv_loop - B (8)
    M 386:4 MultiPeerManager.connect - B (6)
    M 153:4 _RateLimiter._purge - A (5)
    M 264:4 PeerConnection._mark_dead - A (5)
    M 448:4 MultiPeerManager.broadcast - A (5)
    F 59:0 _send_framed - A (4)
    F 100:0 _recv_exactly - A (4)
    C 133:0 _RateLimiter - A (4)
    M 161:4 _RateLimiter.check - A (4)
    C 178:0 PeerConnection - A (4)
    M 203:4 PeerConnection.send - A (4)
    M 460:4 MultiPeerManager.send_to - A (4)
    F 44:0 _validate_frame_length - A (3)
    M 181:4 PeerConnection.__init__ - A (3)
    C 285:0 MultiPeerManager - A (3)
    M 410:4 MultiPeerManager._connect_thread - A (3)
    M 479:4 MultiPeerManager.peer_count - A (3)
    M 488:4 MultiPeerManager.stop_listening - A (3)
    F 78:0 _recv_framed - A (2)
    F 119:0 _is_valid_ip - A (2)
    F 128:0 _is_valid_port - A (2)
    M 319:4 MultiPeerManager.listen_port - A (2)
    M 322:4 MultiPeerManager.listen - A (2)
    M 498:4 MultiPeerManager.disconnect - A (2)
    M 504:4 MultiPeerManager.disconnect_all - A (2)
    M 143:4 _RateLimiter.__init__ - A (1)
    M 218:4 PeerConnection.start_receiver - A (1)
    M 261:4 PeerConnection.close - A (1)
    M 281:4 PeerConnection.is_alive - A (1)
    M 298:4 MultiPeerManager.__init__ - A (1)
    M 434:4 MultiPeerManager._register_socket - A (1)
    M 471:4 MultiPeerManager.get_peer - A (1)
    M 475:4 MultiPeerManager.peer_ids - A (1)
    M 483:4 MultiPeerManager._handle_peer_disconnect - A (1)
```

```
┌──(kali㉿kali)-[/media/sf_Shared/ShushChat_v1]
└─$ radon mi . -s
crypto.py - A (50.63)
gui.py - C (0.00)
main.py - A (88.82)
networking.py - A (38.67)

┌──(kali㉿kali)-[/media/sf_Shared/ShushChat_v1]
└─$ radon raw vulnerable_app.py

┌──(kali㉿kali)-[/media/sf_Shared/ShushChat_v1]
└─$ radon raw .
crypto.py
    LOC: 408
    LLOC: 221
    SLOC: 220
    Comments: 43
    Single comments: 41
    Multi: 69
    Blank: 78
    - Comment Stats
        (C % L): 11%
        (C % S): 20%
        (C + M % L): 27%
gui.py
    LOC: 1420
    LLOC: 1117
    SLOC: 1162
    Comments: 8
    Single comments: 6
    Multi: 0
    Blank: 252
    - Comment Stats
        (C % L): 1%
        (C % S): 1%
        (C + M % L): 1%
main.py
    LOC: 60
    LLOC: 24
    SLOC: 27
    Comments: 1
    Single comments: 0
    Multi: 21
    Blank: 12
    - Comment Stats
        (C % L): 2%
        (C % S): 4%
        (C + M % L): 37%
networking.py
    LOC: 509
    LLOC: 330
```

```
    - Comment Stats
        (C % L): 2%
        (C % S): 4%
        (C + M % L): 37%
networking.py
    LOC: 509
    LLOC: 330
    SLOC: 343
    Comments: 37
    Single comments: 29
    Multi: 57
    Blank: 80
    - Comment Stats
        (C % L): 7%
        (C % S): 11%
        (C + M % L): 18%
```

- **After:**

```
┌──(kali㉿kali)-[/media/sf_Shared/ShushChat_OBF/ShushChat_v1_Final/ShushChat_v1]
└─$ radon cc -s .
crypto.py
    M 210:4 Session._parse_and_verify - B (10)
    M 89:4 KeyPair.derive_shared_key - A (5)
    M 406:4 GroupSession.mark_seen - A (5)
    C 148:0 Session - A (4)
    F 296:0 peer_fingerprint_from_bytes - A (3)
    C 70:0 KeyPair - A (3)
    M 161:4 Session.__init__ - A (3)
    M 178:4 Session.encrypt - A (3)
    M 245:4 Session._check_nonce - A (3)
    M 256:4 Session.decrypt - A (3)
    C 309:0 GroupSession - A (3)
    M 334:4 GroupSession.add_peer - A (3)
    M 400:4 GroupSession.is_duplicate - A (3)
    F 55:0 _zero_bytes - A (2)
    M 84:4 KeyPair.fingerprint - A (2)
    C 129:0 IdentityKey - A (2)
    M 357:4 GroupSession.remove_peer - A (2)
    M 369:4 GroupSession.peer_fingerprint - A (2)
    M 382:4 GroupSession.encrypt - A (2)
    M 390:4 GroupSession.decrypt - A (2)
    M 420:4 GroupSession.clear_all - A (2)
    C 429:0 _CryptoObfuscatorMeta - A (2)
    C 434:0 _DummyCipher - A (2)
    M 73:4 KeyPair.__init__ - A (1)
    M 77:4 KeyPair.public_key_bytes - A (1)
    M 124:4 KeyPair.clear - A (1)
    M 131:4 IdentityKey.__init__ - A (1)
    M 135:4 IdentityKey.public_bytes - A (1)
    M 141:4 IdentityKey.sign - A (1)
    M 278:4 Session._hmac - A (1)
    M 283:4 Session.clear - A (1)
    M 325:4 GroupSession.__init__ - A (1)
    M 364:4 GroupSession.has_peer - A (1)
    M 375:4 GroupSession.active_peer_ids - A (1)
    M 430:4 _CryptoObfuscatorMeta.__new__ - A (1)
    M 435:4 _DummyCipher.__init__ - A (1)
    M 437:4 _DummyCipher.do_nothing - A (1)
gui.py
    M 835:4 ShushChatApp._handle_handshake - D (22)
    M 962:4 ShushChatApp._handle_encrypted - C (15)
    M 1038:4 ShushChatApp._rx_chat - C (15)
    M 918:4 ShushChatApp._rx_peer_list - C (14)
    M 503:4 ShushChatApp._refresh_room_list - B (10)
    M 1094:4 ShushChatApp._rx_room_hello - B (10)
    M 1431:4 _RoomDialog._submit - B (10)
    M 539:4 ShushChatApp._switch_room - B (9)
    M 1274:4 ShushChatApp._rx_conn_req - B (8)
    C 1363:0 _RoomDialog - B (8)
    M 1364:4 _RoomDialog.__init__ - B (8)
```

```
    M 776:4 ShushChatApp._send_handshake - A (1)
    M 788:4 ShushChatApp._on_raw_message - A (1)
    M 809:4 ShushChatApp._on_peer_disconnected - A (1)
    M 827:4 ShushChatApp._on_network_error - A (1)
    M 1164:4 ShushChatApp._set_conn_badge - A (1)
    M 1174:4 ShushChatApp._set_status - A (1)
    M 1177:4 ShushChatApp._unlock_input - A (1)
    M 1183:4 ShushChatApp._lock_input - A (1)
    M 1206:4 ShushChatApp._now - A (1)
    M 1241:4 ShushChatApp._add_err_to_active - A (1)
main.py
    F 47:0 main - A (2)
networking.py
    M 340:4 MultiPeerManager._listen_loop - B (9)
    M 232:4 PeerConnection._recv_loop - B (8)
    M 393:4 MultiPeerManager.connect - B (6)
    M 158:4 _RateLimiter._purge - A (5)
    M 271:4 PeerConnection._mark_dead - A (5)
    M 456:4 MultiPeerManager.broadcast - A (5)
    F 61:0 _send_framed - A (4)
    F 102:0 _recv_exactly - A (4)
    F 519:0 _obfuscate_junk_code_293847 - A (4)
    C 138:0 _RateLimiter - A (4)
    M 166:4 _RateLimiter.check - A (4)
    C 183:0 PeerConnection - A (4)
    M 210:4 PeerConnection.send - A (4)
    M 468:4 MultiPeerManager.send_to - A (4)
    F 46:0 _validate_frame_length - A (3)
    M 186:4 PeerConnection.__init__ - A (3)
    C 292:0 MultiPeerManager - A (3)
    M 417:4 MultiPeerManager._connect_thread - A (3)
    M 487:4 MultiPeerManager.peer_count - A (3)
    M 496:4 MultiPeerManager.stop_listening - A (3)
    F 80:0 _recv_framed - A (2)
    F 121:0 _is_valid_ip - A (2)
    F 130:0 _is_valid_port - A (2)
    M 326:4 MultiPeerManager.listen_port - A (2)
    M 329:4 MultiPeerManager.listen - A (2)
    M 506:4 MultiPeerManager.disconnect - A (2)
    M 512:4 MultiPeerManager.disconnect_all - A (2)
    M 148:4 _RateLimiter.__init__ - A (1)
    M 225:4 PeerConnection.start_receiver - A (1)
    M 268:4 PeerConnection.close - A (1)
    M 288:4 PeerConnection.is_alive - A (1)
    M 305:4 MultiPeerManager.__init__ - A (1)
    M 441:4 MultiPeerManager._register_socket - A (1)
    M 479:4 MultiPeerManager.get_peer - A (1)
    M 483:4 MultiPeerManager.peer_ids - A (1)
    M 491:4 MultiPeerManager._handle_peer_disconnect - A (1)
```

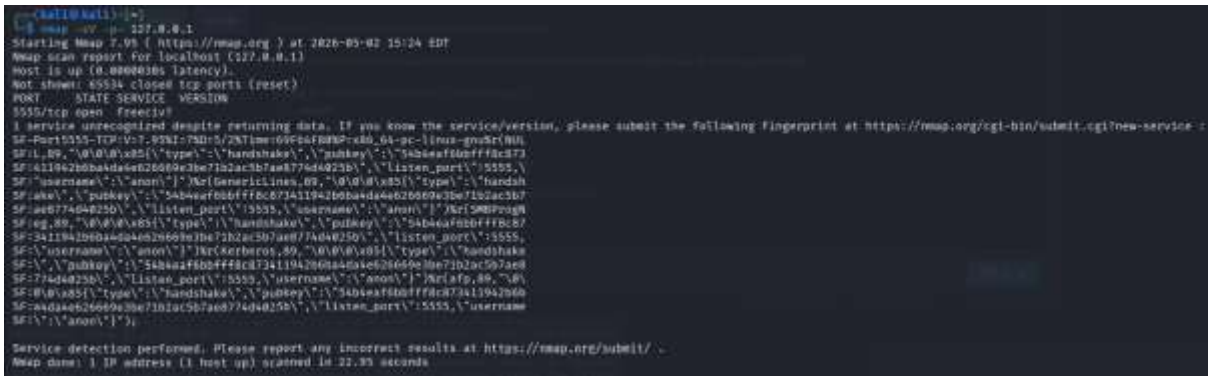
```
┌──(kali㉿kali)-[/media/sf_Shared/ShushChat_OBF/ShushChat_v1_Final/ShushChat_v1]
└─$ radon mi -s .
crypto.py - A (47.34)
gui.py - C (0.00)
main.py - A (88.82)
networking.py - A (37.24)
```

```
└─$ radon raw .
crypto.py
    LOC: 438
    LLOC: 256
    SLOC: 243
    Comments: 37
    Single comments: 45
    Multi: 69
    Blank: 81
    - Comment Stats
        (C % L): 8%
        (C % S): 15%
        (C + M % L): 24%
gui.py
    LOC: 1449
    LLOC: 1141
    SLOC: 1188
    Comments: 7
    Single comments: 5
    Multi: 0
    Blank: 256
    - Comment Stats
        (C % L): 0%
        (C % S): 1%
        (C + M % L): 0%
main.py
    LOC: 60
    LLOC: 24
    SLOC: 27
    Comments: 1
    Single comments: 0
    Multi: 21
    Blank: 12
    - Comment Stats
        (C % L): 2%
        (C % S): 4%
        (C + M % L): 37%
networking.py
    LOC: 525
    LLOC: 344
    SLOC: 357
    Comments: 38
    Single comments: 30
    Multi: 57
    Blank: 81
    - Comment Stats
        (C % L): 7%
        (C % S): 11%
        (C + M % L): 18%
```

# Dynamic Analysis

## NMAP:

- **Before:**

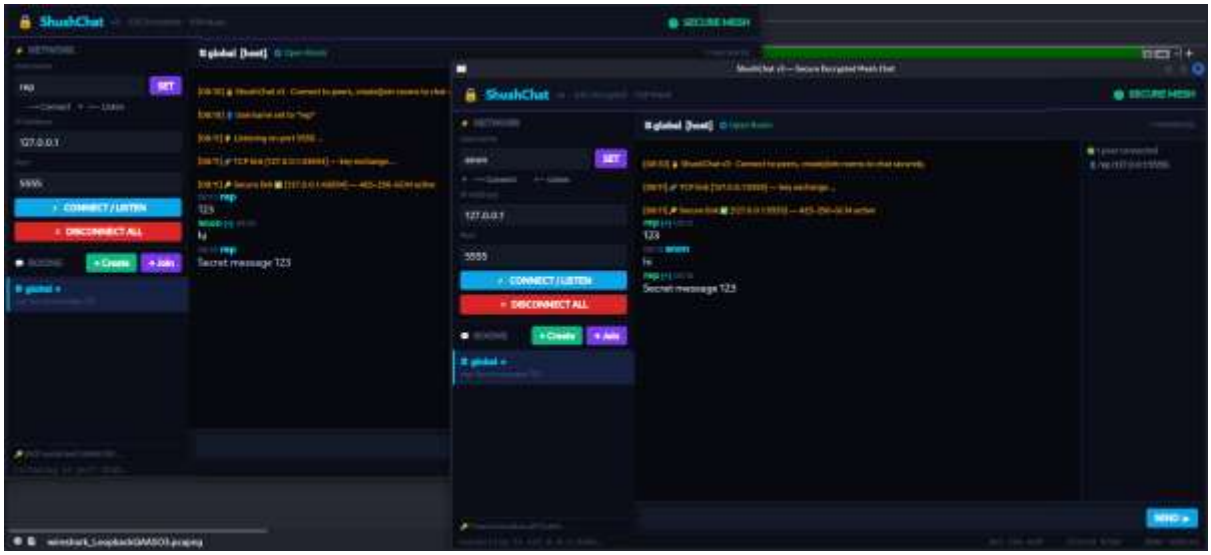


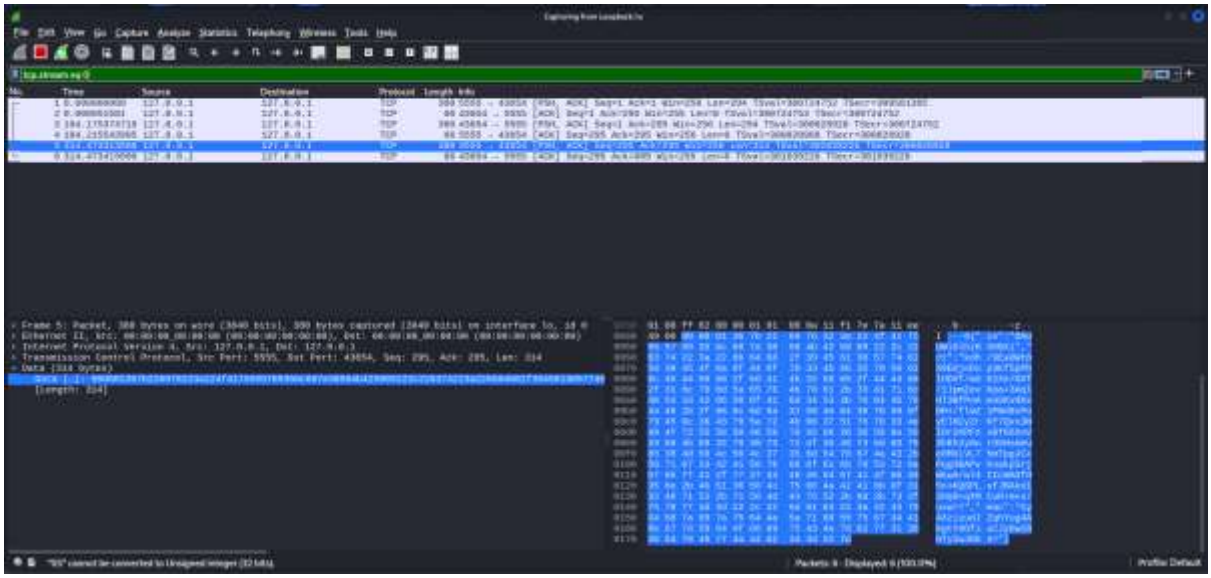
- **After:**

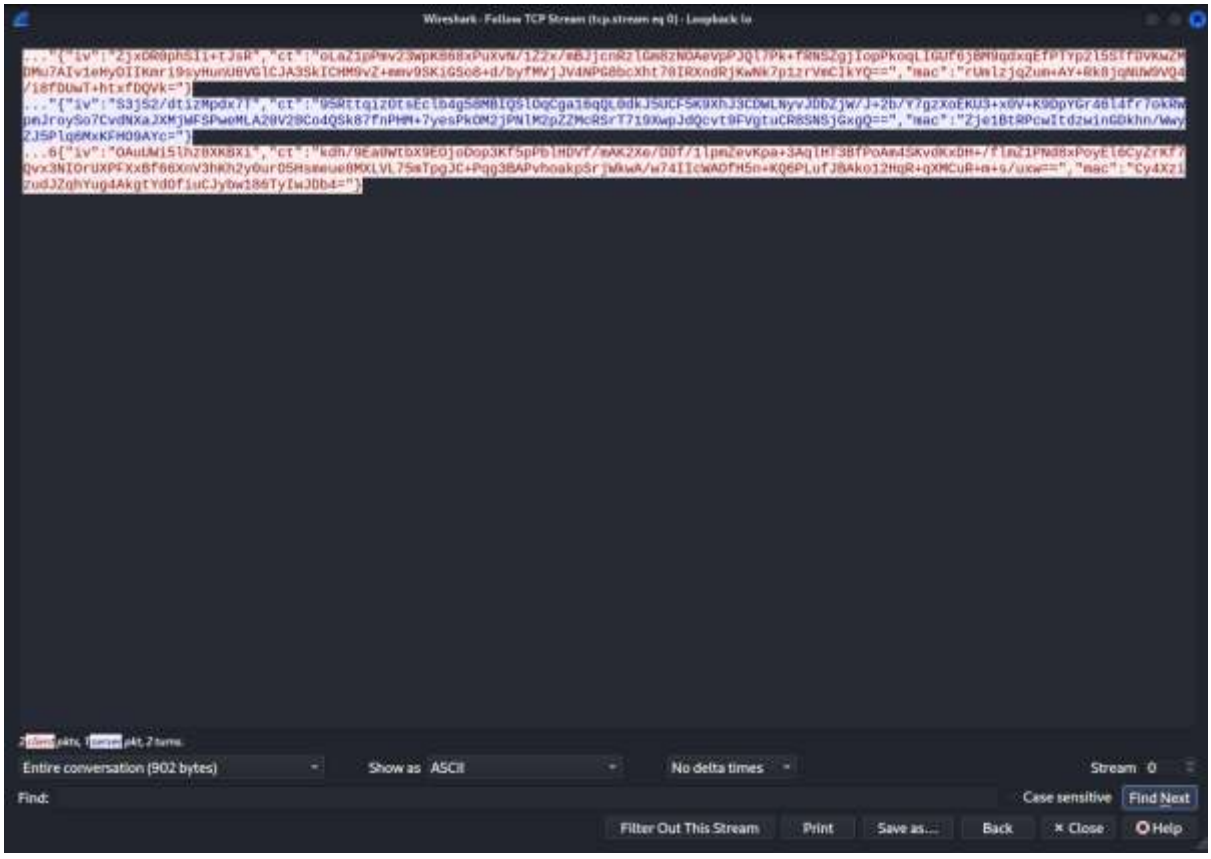
```
┌──(kali㉿kali)-[~]
└─$ nmap -sV -p- 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-05 07:37 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 65534 closed tcp ports (reset)
PORT     STATE SERVICE  VERSION
5555/tcp open  freeciv?

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.36 seconds
```

## WireShark:

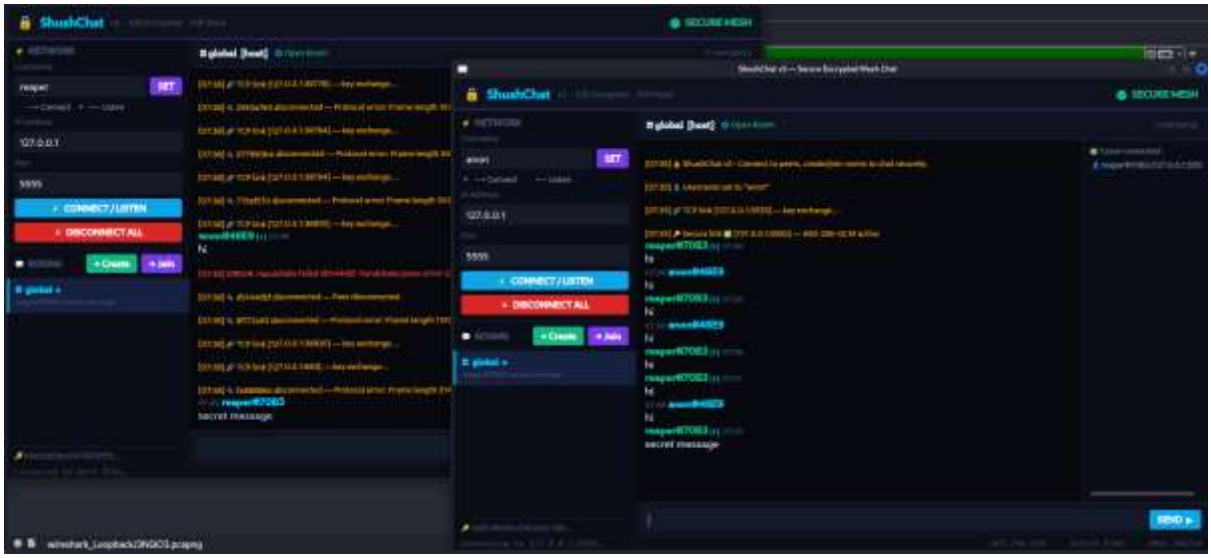
- **Before:**

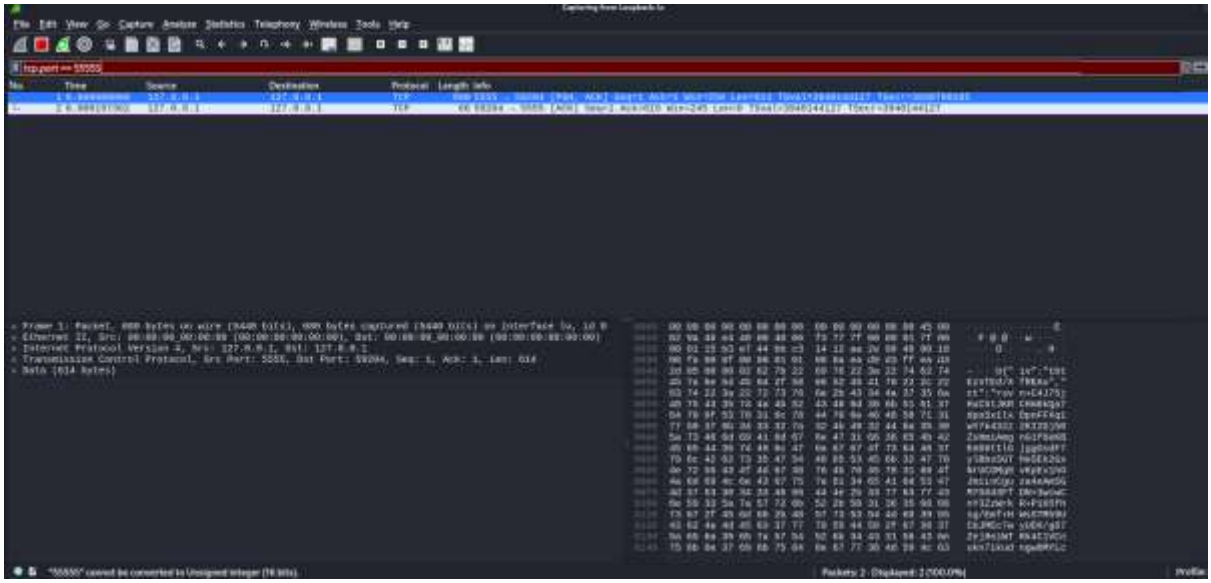





Wireshark · Follow TCP Stream (tcp.stream eq 0) · Loopback: lo
Entire conversation (902 bytes)
Show as ASCII
No delta times
Stream 0
Find:
Case sensitive
Find Next
Filter Out This Stream
Print
Save as…
Back
× Close
Help

- **After:**



...b{"iv":"EbtEznTEd/XfMEAx","ct":"rsvn+C4J75jHuC5tJKRCHm8kQa7dpo5x1lxOpnFFXq1wY7k432z2K12Dj58Z5Hm1AmgnG1f6eKBEeD6tIlGjggOsdF7yiBbs5GlHe
GEk2GxNrUCOMg0vKpEx1hOJmiLnCguza4eAmSGM7S043FfDN+3wowCnY3ZzWrkR+P165fhsg/EmF+HWsSTMhRUCbJMEc7wyUDX/g67Zej9ezWTRk4C1VCnukn7ikudngwMMYLc/5
IeeMOxha7XxKsXqhGZeeEfburMx+r3FQwH2UcQBj7fLlnekqyOLOZTOjeb2jSY1nhG4Wrq4fzTRxwGuBO7FBJSc189SzP+M1D0NYIef5QWG3hvkqCoM3s7bcsS59sIl5qoLqVVZl
jUC/7xmcwT8wsykI2p/qNBBq3cl3hPDsw9OMAODGIKZTsXAPLYf8YilxJ8JcGlFiSWx2cKWwdaNElfmhwHA6UgMboHjPPJbZNAKtOXFUlo+E9IBRVXnCoMvjA0QHLriXBY2tOpwb
QQbPPaLPusa2Q==","mac":"UNuvlFowJ3bn224Y2V013H808wzUuvuzWkCPziEj3oE="}

0 client pkt, 0 server pkts, 0 turns.

Entire conversation (614 bytes) | Show as ASCII | No delta times | Stream 0

Find: | Case sensitive | Find Next

Filter Out This Stream | Print | Save as... | Back | × Close | Help